

Name: \_\_\_\_\_

# Solving Radical Equations

*Advanced Notes*

## Day 1: Equations That Create Quadratics (Extraneous Solutions)

### Process

**Step 1:** Isolate the radical on one side.**Step 2:** Square both sides. (Note: Squaring a binomial creates a quadratic!)**Step 3:** Solve the resulting quadratic equation (set = 0, factor or use quadratic formula).**Step 4:** Check BOTH solutions in the original equation for extraneous solutions.

### Key Concept: Extraneous Solutions

When you square both sides of an equation, you may introduce solutions that don't work in the original equation. These are called ***extraneous solutions***. You **MUST** check all answers!

**Example 1** Solve the equation. Check for extraneous solutions.

**A.**  $\sqrt{2x} = x - 4$

**C.**  $\sqrt{x+6} = x - 6$

**B.**  $\sqrt{2x-5} = x - 10$

**Connection to Rational Exponents:** Remember that  $\sqrt{x} = x^{1/2}$ . This means radical equations can also be written using rational (fractional) exponents!

Examples:  $\sqrt{x+3} = x^{1/2} + 3$  and  $x^{1/2} = 5$  is the same as  $\sqrt{x} = 5$

To solve equations with rational exponents, use the SAME PROCESS!

### Process for Rational Exponents

**Step 1:** Make sure the term with the rational exponent is alone on one side. **Step 2:** Raise both sides to the reciprocal power (this eliminates the exponent).

- If you have  $x^{1/2}$ , raise both sides to the 2nd power
- If you have  $x^{1/3}$ , raise both sides to the 3rd power

**Step 3:** Solve for  $x$ . **Step 4:** Check your answer in the original equation.

**Example 2** Solve the equation. Check your solution.

A.  $(x - 4)^{1/2} = 6$

B.  $(3x + 5)^{1/2} = 8$

**Day 2: Two Radicals – Squaring Twice****Process (When radicals are NOT isolated from each other)**

**Step 1:** Isolate ONE radical on one side of the equation.

**Step 2:** Square both sides. (A radical will remain on one side.)

**Step 3:** Isolate the remaining radical.

**Step 4:** Square both sides again.

**Step 5:** Solve and check for extraneous solutions.

**When to Square Twice:** If one radical is NOT isolated (there's a constant or another radical with it on the same side), you'll need to square twice.

**Example 2** Solve the equation. Check for extraneous solutions.

**A.**  $\sqrt{x-5} + 1 = \sqrt{x+2}$

**C.**  $\sqrt{x+4} - 2 = (x+8)^{\frac{1}{2}}$

(This one has no solution – can you see why?)

**B.**  $\sqrt{x+4} - 2 = \sqrt{x-4}$

**Day 3: Mixed Practice & Applications****Strategy Summary: How to Identify the Type**

- One radical = expression with  $x \rightarrow$  Square once, get quadratic, check for extraneous
- Two radicals, one on each side (isolated)  $\rightarrow$  Square once, solve linear
- Two radicals, NOT isolated  $\rightarrow$  Square twice, check for extraneous

**Example 3** Identify the type, then solve. Check for extraneous solutions.

A.  $(5x + 24)^{\frac{1}{2}} = x$

Type: \_\_\_\_\_

B.  $\sqrt{3x - 2} = \sqrt{x + 6}$

Type: \_\_\_\_\_

C.  $\sqrt{2x + 3} - \sqrt{x - 2} = 2$

Type: \_\_\_\_\_

## Radical Equations Application

**Example 4** The period  $T$  (in seconds) of a pendulum is given by:

$$T = 2\pi\sqrt{\frac{L}{32}}$$

where  $L$  is the length of the pendulum in feet. A grandfather clock has a pendulum with a period of 2 seconds. Find the length of the pendulum.